Average gate fidelity at $\Omega_p = 2\pi \cdot 40$ MHz

Option A: Rb $66D_{\frac{5}{2}}$ + Cs $76D_{\frac{3}{2}}$ Förster pair,
$a = 5.00$ µm, ARC-computed lifetimes ($T = 300$ K), $\Omega_R = 2.9\ \Omega_p$

TriQG `examples/Average_fidelity_Omega_p` · compiled 2026-04-11

TL;DR. Target probe amplitude set to $\Omega_p = 2\pi \cdot 40$ MHz. The super-Gaussian pulse window T_f and width σ are rescaled analytically so that the effective two-photon pulse area ($\pi/4$) and the pulse edge shape (edge/peak $\approx 2.6\%$) are both exactly preserved. Results:

- **OR gate:** $\overline{F}_{\mathrm{OR}} = \mathbf{0.992301}$, infidelity $\mathbf{7.70 \times 10^{-3}}$.
- **CCX gate:** $\overline{F}_{\mathrm{CCX}} = \mathbf{0.996620}$, infidelity $\mathbf{3.38 \times 10^{-3}}$.

At this Ω_p, Raman scattering off $|7P_{\frac{3}{2}}\rangle$ is numerically negligible ($P(|P\rangle) < 10^{-4}$ on every branch), because the in-pulse $|P\rangle$ population scales as $\left(\Omega_p/\Delta\right)^2 = (40/500)^2 = 6.4 \times 10^{-3}$. The OR gate infidelity is dominated by the doubly-blockaded $|1, 1, *\rangle$ branch (1.5×10^{-2} per input), which absorbs the accumulated V_{cc} phase over the 468.75 ns target window.

1 Scope

This run evaluates the Option A Rydberg pair (Rb $66D_{\frac{5}{2}}$ + Cs $76D_{\frac{3}{2}}$, Ireland, Pritchard & Shaffer 2024 Table I row 1) at lattice spacing $a = 5.00$ µm, with the following pulse-side choices:

Quantity	symbol	value
Cs control Rabi	Ω_c	$2\pi \cdot 50$ MHz
Rb target probe Rabi	Ω_p	$\mathbf{2\pi \cdot 40}$ **MHz**
Rb upper-leg coupling	$\Omega_R = 2.9\ \Omega_p$	$2\pi \cdot 116$ MHz
Cs CCX control Rabi	Ω_{cc}	$2\pi \cdot 100$ MHz
Rb CCX target Rabi	Ω_t	$2\pi \cdot 50$ MHz
Two-photon detuning (OR)	Δ	$2\pi \cdot 500$ MHz
Control π-pulse (OR)	$T_c = \pi/\Omega_c$	10.000 ns
Target half-window (OR)	T_f	**234.375 ns**
Super-Gaussian width (OR)	σ	**0.006755829 µs**
Total OR gate time	$2T_c + 2T_f$	**488.75 ns**
Control π-pulse (CCX)	$T_{\mathrm{cc}} = \pi/\Omega_{\mathrm{cc}}$	5.000 ns
Target sub-pulse (CCX)	$T_t = \pi/\Omega_t$	10.000 ns
Total CCX gate time	$2T_{\mathrm{cc}} + 3T_t$	40.000 ns

Table 1: Pulse and detuning parameters.

2 Target pulse reshaping

The super-Gaussian target pulse used by `or_average_gate_fid_gaussian.py` has the form

$$\Omega_p(t) = \frac{A}{2} \cdot \exp\left[-\left(\frac{(t-t_c)^3}{\sigma}\right)^2\right],$$

with t_c the pulse center and A the amplitude. Substituting $w = (t - t_c)/\sigma^{\frac{1}{3}}$ in the two-photon pulse-area integral yields

$$\text{area} = A^2 \frac{\sigma^{\frac{1}{3}}}{8\Delta} \cdot J\left(T_f/\sigma^{\frac{1}{3}}\right), \quad J(x) = \int_{-x}^{x} e^{-2w^6}\,\mathrm{d}w.$$

The edge-to-peak ratio is $\exp\left[-\left(T_f^3/\sigma\right)^2\right]$, which also depends only on the ratio $T_f/\sigma^{\frac{1}{3}}$.

Two constraints fix T_f and σ once A is chosen:

1. **Effective two-photon pulse area** stays at $\pi/4$.
2. **Pulse edge shape** (edge/peak ratio) stays fixed, so that the super-Gaussian starts and ends smoothly instead of acquiring a hard cutoff at $t = T_c$ and $t = T_c + 2T_f$.

Holding $x \equiv T_f/\sigma^{\frac{1}{3}}$ fixed freezes the pulse shape and freezes J; the area constraint then reduces to $A^2\sigma^{\frac{1}{3}} = \text{const}$. Combined with the shape constraint,

$$\sigma \propto A^{-6}, \qquad T_f \propto A^{-2}.$$

At $A = 2\pi \cdot 40$ MHz, the pulse parameters used in this run are

$$T_f = 0.234375 \ \ \mu\text{s}, \quad \sigma = 0.006755829 \ \ \mu\text{s},$$

which give the pulse invariants

Invariant	**formula**	**value**
Shape ratio	$x = T_f/\sigma^{\frac{1}{3}}$	1.2398
Edge / peak	$\exp\left[-\left(T_f^3/\sigma\right)^2\right]$	2.647×10^{-2}
Two-photon pulse area	$\int \Omega_p^2/(2\Delta)\,\mathrm{d}t$	$0.785353 \approx \pi/4$

Table 2: Numerical verification of the super-Gaussian invariants at the chosen $\left(T_f, \sigma\right)$. The "area" line is computed by `compute_pulse_area(omega_gaussian, delta, T_c, T_c + 2 T_f, args)`; the others are closed-form.

The pulse is therefore an order-6 super-Gaussian with a flat top, soft edges at $\approx 2.6\%$ of peak, and a pulse area calibrated to $\frac{\pi}{4}$ to six decimal places. The total OR gate time $2T_c + 2T_f = 488.75$ ns is the direct consequence of stretching T_f when A is lowered; this is the only "free" consequence of the reshaping, and drives the cost analysis in §5.

3 Interaction coefficients at $a = 5.00$ μm

Derived blockade strengths for the Option A pair at $\theta = 90°$ (quantization axis perpendicular to the array plane). V_{ct} is the Rb–Cs dipole–dipole Förster coefficient at the data–ancilla distance $r_{\text{DA}} = a/\sqrt{2}$; V_{cc} is the Cs–Cs van der Waals interaction at the ancilla–ancilla distance $r_{\text{AA}} = a\sqrt{2}$.

Quantity	**value**
Lattice spacing	$a = 5.00$ µm
Nearest data–ancilla distance	$r_{\mathrm{DA}} = 3.5355$ µm
Nearest ancilla–ancilla distance	$r_{\mathrm{AA}} = 7.0711$ µm
Förster pair	Rb $66D_{\frac{5}{2}}$ + Cs $76D_{\frac{3}{2}}$
Förster channel	$\lvert$Rb $67P_{\frac{3}{2}}$; Cs $74F_{\frac{5}{2}}\rangle$
Effective Förster $\tilde{C}_3$	22.84 GHz·µm³
Cs–Cs van der Waals C_6^{CsCs}	−692.9 GHz·µm⁶
$V_{\mathrm{ct}}/(2\pi)$	+516.810 MHz
$V_{cc}/(2\pi)$	−5.543 MHz
Selectivity $V_{\mathrm{ct}}/\ \lvert V_{cc}\rvert$	93.2
$\boldsymbol{V_{\mathrm{ct}}/\Delta}$ **(OR)**	**1.034**
V_{ct}/Ω_p (OR)	12.92
V_{ct}/Ω_t (CCX)	10.34
$\lvert V_{cc}\rvert\ /\Omega_c$	0.111
$\lvert V_{cc}\rvert\ /\Omega_{\mathrm{cc}}$	0.055
Blockade fidelity P_{1r}	0.9997

Table 3: Interaction strengths and dimensionless ratios at $a = 5.00$ µm. Bolded row is the tight margin that dominates single-blockade OR-gate errors in §5.

Source of the $\widetilde{C}_3$ / C_6^{CsCs} values: B. J. Ireland, J. D. Pritchard, J. P. Shaffer, **"Interspecies Förster resonances of Rb–Cs Rydberg d-states for enhanced multi-qubit gate fidelities"**, Phys. Rev. Research **6**, 013293 (2024), arXiv:2401.02308, Table I row 1. Values tabulated locally in `reference/energy_level_inter.md`.

4 ARC-computed lifetimes at $T = 300$ K

Lifetimes for the two Rydberg states were computed with ARC 3.10.2 using `atom.getStateLifetime(n, l, j, temperature=300, includeLevelsUpTo=n+30)`. ARC combines the radiative Einstein-A sum with the Beterov et al. 2009 blackbody-radiation-induced depopulation formula.

State	τ_{rad} **(µs)**	τ_{BBR} **(µs)**	τ_{total} **(µs)**
Rb $\lvert 66D_{\frac{5}{2}}\rangle$	294.31	248.96	**134.87**
Cs $\lvert 76D_{\frac{3}{2}}\rangle$	260.15	316.22	**142.73**
Rb $\lvert 7P_{\frac{3}{2}}\rangle$ (paper)	—	—	**0.131**

Table 4: ARC-computed BBR-included lifetimes at $T = 300$ K. τ_{rad} is the radiative component (at $T = 0$); τ_{BBR} is the BBR-induced depopulation time; $\tau_{\mathrm{total}}^{-1} = \tau_{\mathrm{rad}}^{-1} + \tau_{\mathrm{BBR}}^{-1}$. The intermediate $\lvert 7P_{\frac{3}{2}}\rangle$ state retains the paper value $\tau_P = 0.131$ µs.

Sources:

1. N. Šibalić, J. D. Pritchard, C. S. Adams, K. J. Weatherill, **"ARC: An open-source library for calculating properties of alkali Rydberg atoms"**, Comput. Phys. Commun. **220**, 319 (2017), arXiv:1612.05529. Package homepage: https://arc-alkali-rydberg-calculator.readthedocs.io.
2. I. I. Beterov, I. I. Ryabtsev, D. B. Tretyakov, V. M. Entin, **"Quasiclassical calculations of blackbody-radiation-induced depopulation rates and effective lifetimes of Rydberg $\boldsymbol{nS}$, $\boldsymbol{nP}$, and $\boldsymbol{nD}$ alkali-metal atoms with $\boldsymbol{n \le 80}$"**, Phys. Rev. A **79**, 052504 (2009), arXiv:0902.4995.

5 OR gate results

#	Input	F_k
1	$\|0,0,A\rangle$	0.995384
2	$\|0,0,B\rangle$	0.995384
3	$\|0,1,A\rangle$	0.994474
4	$\|0,1,B\rangle$	0.994474
5	$\|1,0,A\rangle$	0.994474
6	$\|1,0,B\rangle$	0.994474
7	$\|1,1,A\rangle$	0.984872
8	$\|1,1,B\rangle$	0.984872

Table 5: OR gate per-input state fidelities at $\Omega_p = 2\pi \cdot 40$ MHz. All 8 computational basis states are simulated via `mesolve` with the collapse operators built from the lifetimes in §4.

OR gate average fidelity:

$$\overline{F}_{\mathrm{OR}} = \frac{1}{8}\sum_{k=1}^{8} F_k = \mathbf{0.992301}, \quad 1 - \overline{F}_{\mathrm{OR}} = \mathbf{7.70 \times 10^{-3}}.$$

5.1 Diagnostic: target-level populations

For each input, the script reports the target atom's projection onto the four relevant Rb levels: the logical $|A\rangle$, $|B\rangle$ ground states, the intermediate $|P\rangle = |7P_{\frac{3}{2}}\rangle$, and the Rydberg $|R\rangle = |66D_{\frac{5}{2}}\rangle$.

#	Input	$P(\|A\rangle)$	$P(\|B\rangle)$	$P(\|P\rangle)$	$P(\|R\rangle)$
1	$\|0,0,A\rangle$	0.9954	0.0032	0.0000	0.0014
2	$\|0,0,B\rangle$	0.0032	0.9954	0.0000	0.0014
3	$\|0,1,A\rangle$	0.0022	0.9945	0.0000	0.0001
4	$\|0,1,B\rangle$	0.9945	0.0022	0.0000	0.0001
5	$\|1,0,A\rangle$	0.0022	0.9945	0.0000	0.0001
6	$\|1,0,B\rangle$	0.9945	0.0022	0.0000	0.0001
7	$\|1,1,A\rangle$	0.0019	0.9849	0.0000	0.0000
8	$\|1,1,B\rangle$	0.9849	0.0019	0.0000	0.0000

Table 6: OR gate diagnostic: target-level populations at the end of the gate. Expected (ideal) populations: $|0,0,*\rangle$ branches should return to the input state; every other branch should flip $A \leftrightarrow B$.

Per-branch infidelity budget.

- **$|0,0,*\rangle$ branches ($1 - F_k = 4.6 \times 10^{-3}$ each).** Neither control is in $|r\rangle$, so there is no blockade. The target executes the full $|A\rangle \to |P\rangle \to |R\rangle \to |P\rangle \to |A\rangle$ cycle and should return unchanged. The 0.0032 miscount is the imperfect-cycle residual, and the 0.0014 population stuck in $|R\rangle$ is EIT-dark-state leakage at the mixing angle set by $\Omega_R/\Omega_p = 2.9$. Notably, $P(|P\rangle) < 10^{-4}$: Raman scattering off the short-lived intermediate state has essentially turned off at this Ω_p.
- **$|0,1,*\rangle$ / $|1,0,*\rangle$ branches ($1 - F_k = 5.5 \times 10^{-3}$ each).** One control is in $|r\rangle$, so the target $|R\rangle$ level is shifted by $V_{\mathrm{ct}}/(2\pi) = 516.81$ MHz. This is close to but larger than $\Delta/(2\pi) = 500$ MHz, giving $V_{\mathrm{ct}}/\Delta = 1.034$. The 0.9945 flipped-target population shows the blockade is working, but the tight margin lets $\approx 0.22\%$ of the amplitude leak back through the off-resonant two-photon path.

The 0.0001 residual $|R\rangle$ population confirms the single-control blockade is suppressing $|R\rangle$ excursion correctly.

- **$|1,1,*\rangle$ branches ($1-F_k = 15.1\times10^{-3}$ each).** Both controls are in $|r\rangle$, so the blockade shift is twice as large, and an additional $V_{cc}/(2\pi) = -5.543$ MHz phase accumulates over the $2T_f = 468.75$ ns target window. The 0.0019 residual "old" target population is even smaller than on the single-blockade branches, but the accumulated V_{cc} phase rotates the doubly-blockaded amplitude by $\approx |V_{cc}|\cdot 2T_f \approx 1.04\pi$, landing on a near-destructive point for the final state alignment. This is the dominant contributor to the OR gate infidelity.

6 CCX gate results

The CCX script (`ccx_average_gate_fidelity.py`) uses independent square pulses $\Omega_{\rm cc}$ and Ω_t and does **not** depend on Ω_p, so the CCX numbers are invariant under the target-probe rescaling of §2. They are included here for a complete picture of the two Option-A gates on this template.

#	Input	F_k
1	$\|0,0,A\rangle$	0.992175
2	$\|0,0,B\rangle$	0.991581
3	$\|0,1,A\rangle$	0.996745
4	$\|0,1,B\rangle$	0.997920
5	$\|1,0,A\rangle$	0.996745
6	$\|1,0,B\rangle$	0.997920
7	$\|1,1,A\rangle$	0.999937
8	$\|1,1,B\rangle$	0.999937

Table 7: CCX gate per-input state fidelities. Pulse parameters: $\Omega_{\rm cc} = 2\pi\cdot 100$ MHz, $\Omega_t = 2\pi\cdot 50$ MHz, $T_{\rm cc} = 5$ ns, $T_t = 10$ ns, total gate time 40 ns.

CCX gate average fidelity:

$$\overline{F}_{\rm CCX} = \frac{1}{8}\sum_{k=1}^{8} F_k = \mathbf{0.996620}, \quad 1-\overline{F}_{\rm CCX} = \mathbf{3.38\times10^{-3}}.$$

The CCX gate is dominated by the $|0,0,*\rangle$ branch, which loses $\approx 8\times10^{-3}$ per input from the V_{cc} phase during the Cs control π-pulse window. The $|1,1,*\rangle$ branch is essentially perfect ($F_k = 0.999937$), and the single-blockade branches each contribute about 2.7×10^{-3} from the $V_{\rm ct}/\Delta$ margin (same underlying mechanism as on the OR gate's single-blockade branches, even though CCX uses square pulses).

7 Reproducibility

Environment (Apple Silicon, Python 3.12, `qutip == 5.2.3`, `arc-alkali-rydberg-calculator == 3.10.2`):

```
source .venv/bin/activate
python examples/Average_fidelity_Omega_p/or_average_gate_fid_gaussian.py
python examples/Average_fidelity_Omega_p/ccx_average_gate_fidelity.py
```

Key parameters in the OR gate script (`or_average_gate_fid_gaussian.py`):

- `omega_c_amp = 2 * np.pi * 50`
- `omega_p_amp = 2 * np.pi * 40.0`
- `omega_R_amp = 2.9 * omega_p_amp` ($\rightarrow 2\pi\cdot 116$ MHz)

- `delta = 2 * np.pi * 500`
- `T_c = np.pi / omega_c_amp` ($\rightarrow$ 10 ns)
- `T_f = 0.15 * (5 / 4) ** 2` ($\rightarrow$ 0.234375 µs = 234.375 ns)
- `sigma = 0.001771 * (5 / 4) ** 6` ($\rightarrow \approx 0.006756$ µs)
- `a_um = 5.0`
- `C3_tilde = 22.84`, `C6_CsCs = -692.9` (Ireland et al. 2024)
- `gamma_r = 1 / 142.73`, `gamma_R = 1 / 134.87` (ARC, 300 K)
- `gamma_P = 1 / 0.131` (paper value retained)

Key parameters in the CCX gate script (`ccx_average_gate_fidelity.py`): unchanged relative to the previous template – $\Omega_{\mathrm{cc}} = 2\pi \cdot 100$ MHz, $\Omega_t = 2\pi \cdot 50$ MHz, $T_{\mathrm{cc}} = \pi/\Omega_{\mathrm{cc}} = 5$ ns, $T_t = \pi/\Omega_t = 10$ ns, same interaction coefficients and lifetimes as the OR gate script.

8 Conclusion

1. **The super-Gaussian scaling law is exact.** Holding the shape ratio $x = T_f/\sigma^{\frac{1}{3}}$ fixed and imposing $A^2\sigma^{\frac{1}{3}} = \mathrm{const}$ uniquely determines $\sigma \propto A^{-6}$ and $T_f \propto A^{-2}$. Numerical verification gives pulse area $= 0.785353 = \pi/4$ to six decimal places and edge/peak ratio $= 2.647 \times 10^{-2}$, matching closed-form expectations bit-for-bit.

2. **OR gate average fidelity:** $\overline{F}_{\mathrm{OR}} = 0.992301$, infidelity 7.70×10^{-3}. The dominant error is on the doubly-blockaded $|1, 1, *\rangle$ branch (1.5×10^{-2} per input), from the accumulated Cs–Cs V_{cc} phase over the now-longer 468.75 ns target window. Single-blockade branches each lose 5.5×10^{-3} from the tight $V_{\mathrm{ct}}/\Delta = 1.034$ margin. The no-blockade $|0, 0, *\rangle$ branch loses 4.6×10^{-3}, split between a 0.0032 imperfect-cycle residual and 0.0014 EIT-dark-state $|R\rangle$ leakage.

3. **Raman scattering is off.** With $\Omega_p = 2\pi \cdot 40$ MHz, the in-pulse $|P\rangle$ population is $\left(\Omega_p/(2\Delta)\right)^2 \approx 1.6 \times 10^{-3}$, so the $|P\rangle$-state loss channel (short $\tau_P = 0.131$ µs lifetime) contributes $< 10^{-4}$ on every branch. The longer target window does **not** cost fidelity through Raman scattering at this Ω_p.

4. **CCX gate average fidelity:** $\overline{F}_{\mathrm{CCX}} = 0.996620$, infidelity 3.38×10^{-3}, unchanged because the CCX pulses do not depend on Ω_p.

5. **Where the infidelity budget points next.** The $|1, 1, *\rangle$ V_{cc} phase is the largest remaining lever on the OR gate; the V_{ct}/Δ margin is the second. Both are properties of the geometry ($a = 5.00$ µm) and the Option A pair, not of Ω_p. Further reductions in Ω_p will stretch T_f (as A^{-2}) and grow the $|1, 1, *\rangle$ V_{cc} phase by the same factor; a sensible next step is to scan Ω_p upward until $V_{cc} \cdot 2T_f$ lands on a constructive-alignment point, rather than lowering it further.